[INCH-POUND]

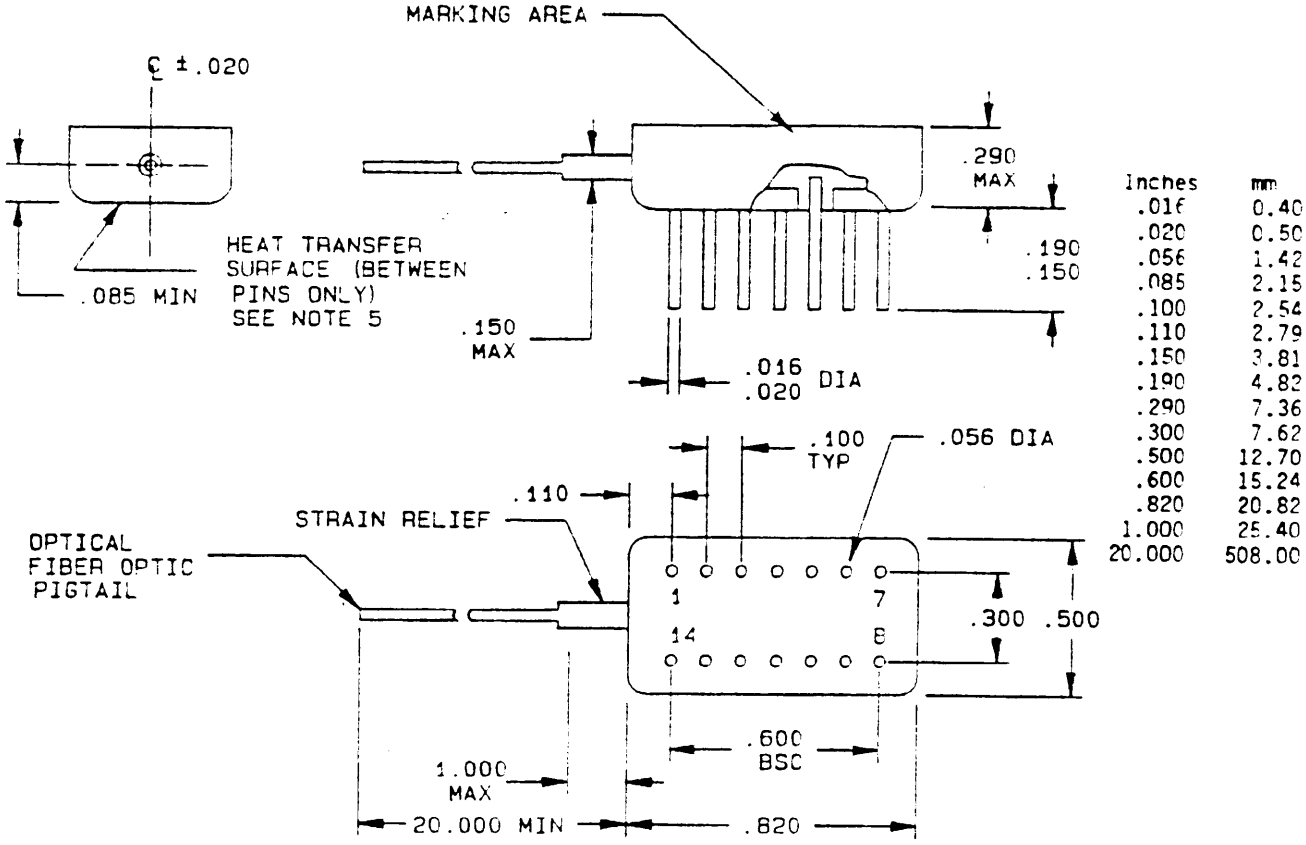
MIL-D-24620/4A
30 June 1989
SUPERSEDING
MIL-S-24620/4(CR)
23 January 1987

MILITARY SPECIFICATION SHEET

DETECTOR, PINFET, FIBER OPTIC, 1100-1600 NANOMETERS, GLASS PIGTAILED, HERMETICALLY SEALED, DUAL IN-LINE PACKAGE (DIP)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-D-24620.



Inches	mm
.016	0.40
.020	0.50
.056	1.42
.085	2.15
.100	2.54
.110	2.79
.150	3.81
.190	4.82
.290	7.36
.300	7.62
.500	12.70
.600	15.24
.820	20.82
1.000	25.40
20.000	508.00

FIGURE 1. PINFET detector - (DIP) package outline and pin connections.

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Pin connections

Pin number	Function
1	Detector bias (-5 V)
2	NC
3	Ground
4	-5 V supply ±0.5 V
5	Ground
6	NC
7	Output (must be externally ac coupled)
8	Ground
9	NC
10	+5 V supply ±0.5 V
11	NC
12	NC
13	AGC control
14	NC

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm).
4. Leads shall be gold (50 microinches minimum) in accordance with MIL-G-45204, type II, grade A, class 1 over nickel (100 microinches minimum) in accordance with QQ-N-290, class 1, grade C or MIL-C-26074.
5. The heat transfer surface (heatsink) shall provide the major heat transfer mechanism to remove heat from the device. Convection and radiation to the environment surrounding the device must not provide significant heat transfer for this device when it is being tested to meet this specification.
6. A yellow band with the phrase "fiber optic" included within it shall be provided on the strain relief of the fiber optic pigtail.

FIGURE 1. PINFET detector - (DIP) package outline and pin connections - Continued

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REQUIREMENTS:

Design and construction: See figure 1.

Pigtail fibers. As specified by DOD-F-49291/1 with the exception that the buffer coating be upjacketed to a diameter of 900 µm ±50 µm. The fiber end termination shall be unconnectorized.

Fiber retention:

Tension: Fiber pigtails shall withstand a minimum tensile load of 5 newtons for a minimum of 30 seconds.

Environmental requirements:

Temperature range:

Operating: -46°C to +85°C.

Storage: -57°C to +85°C.

Nuclear radiation resistance: The nuclear radiation resistance characteristics of this detector are classified and are contained in an appendix to this specification. Application to receive this appendix must be made through the US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ, 07703-5016. Information concerning security clearance classification and "need to know" must be detailed in the request. When measured 15 minutes after the specified irradiation, the detector shall satisfy all the requirements contained herein except as specified in table I.

Electrical and optical requirements: See table II.

Screening: Screening shall be conducted on 100 percent of all devices prior to groups A, B, and C inspections and shall be in accordance with the methods specified in table III. Tests shall be performed in accordance with method 5008 of MIL-STD-883.

Group A inspection: Group A inspection shall consist of the examinations and tests specified in table IV.

Group B inspection: Group B inspection shall consist of the examinations and tests specified in table V.

Group C inspection: Group C inspection shall consist of the examinations and tests specified in table VI.

Part or Identifying Number (PIN): M24620/4-001.

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TABLE I. Post nuclear requirements. 1/

Responsivity R (V/W)	Output noise voltage (V_N) (μV)		Minimum sensitivity S (dBm)
	Temperature	Noise limit	
Minimum 94 percent of table II	-46°C ±2°C	480	-46.2
	+25°C ±2°C	540	-45.7
	+85°C ±2°C	602	-45.2

1/ The detector module shall satisfy all the requirements previously specified except for those parameters stated in this table after nuclear exposure.

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TABLE II. Electrical and optical requirements. 1/

Detector dash number M24620/4-	Optical	Electro-optical						Electrical			
	Spectral sensitivity 2/ λp (nm)	AGC control voltage 3/	Maximum input optical power (dBm)	Minimum bandwidth (MHz)	Bandwidth flatness (dB) 7/	Responsivity R (volts/watt) 7/	Minimum sensitivity S (dBm) 4/	Load resistance (Ω)	Output noise voltage V_N (μV) 5/		Total power consumption (mW) 6/
									Temperature range	Noise limit	
001	1200 to 1600	AGC off	-7.5	dc to 18	+1	55,000 to 120,000	-48.5	≥ 1000 ac coupled	-46°C	285	< 200 for each device
							-48.0		+25°C	320	
		AGC on				16 dB to 30 dB reduction	-47.5		+85°C	355	< 75 from negative supply

1/ Unless otherwise specified, the ambient temperature shall be +25°C ±2°C.

2/ -1 dB responsivity relative to 1300 nm. PIN detector wafer screen. Test five devices from each wafer, one from each corner and from center.

3/ The AGC control voltage shall be -4.5 V dc for full off and -1.5 V dc for full on.

4/ $S = 10 \text{ LOG } \frac{V_N * 1000}{R} + 7.8$.

5/ V_N is the total RMS noise voltage appearing across the load resistor for a measurement bandwidth of 100 kHz to 13 MHz. The low pass characteristics of the noise filter shall be that of a 2-pole Butterworth having a -3 db corner at no less than 13 MHz.

6/ The normal operating voltages shall be applied to the module and the input currents shall be measured. This test shall be performed with the fiber optic pigtail input covered.

7/ The responsivity ratio, expressed in decibels, is 20 LOG (R_2/R_1).

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TABLE III. Screening inspection.

Screen inspection (100%)	MIL-STD-883 method	Details
Internal visual inspection	2017	Ambient
Stabilization bake	1008	24 hours minimum at T = +85°C
Thermal shock (temperature cycling)	1010	Test condition A, steps 1 and 3 shall be 2 hours minimum
Constant acceleration	2001	Y1 orientation only at 5,000 G minimum
Hermetic seal Fine leak rate Gross leak rate	1014	 Test condition A_1, 5×10^{-7} atm cm^3/s maximum leak rate. Test condition C, except that leak indicator fluid shall be maintained at +100°C ±5°C
Steady-state burn-in	1015	Condition C for 160 hours minimum, temperature (heat sink) = +85°C
External visual and mechanical inspection	2009	Ambient
End points: Subgroup 1 Group A		 $\Delta R \leq 20\%$ $\Delta V_N \leq 20\%$

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TABLE IV. Group A inspection. 1/

Inspection	Details	LTPD
Subgroup 1 2/	+25°C	
Module responsivity		5
Sensitivity		5
Output noise		5
Bandwidth and flatness		5
Power consumption		5
Subgroup 2	-46°C	
Module responsivity		10
Output noise		10
Power consumption		10
Sensitivity		10
Bandwidth and flatness		10
Subgroup 3	+85°C	
Module responsivity		10
Output noise		10
Power consumption		10
Sensitivity		10
Bandwidth and flatness		10
Subgroup 4	+25°C	
Maximum optical power		10
Power supply 2/		15

1/ All measurements except for spectral sensitivity shall be made at a wavelength of 1290 nm ±20 nm.

2/ The modules shall meet the responsivity and output noise requirements of subgroup 1 for the following conditions of power supply voltages:

Condition	Positive supply	Negative supply
A	+5.50	-5.50
B	+5.50	-4.50
C	+4.50	-5.50
D	+4.50	-4.50

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TABLE V. Group B inspection.

Inspection	Method	MIL-STD-883 details	LTPD
Subgroup 1 1/			
Solderability 2/	2003		15
Lead integrity	2004	Test condition C_1	15
Resistance to solvents	2015		15
Subgroup 2 1/			
Internal visual and mechanical	2014		15
Bond strength	2011		10
Die shear strength	2019		15
Subgroup 3			
Physical dimensions	2015		20
Fiber retention			10
Seal: Fine Gross	1014	Test condition A_1 for fine leaks (leak testing 2 hours after pressurization is acceptable); test condition C_1 for gross leaks, except that leak indicator fluid shall be maintained at +100°C ±5°C.	10
End points: Subgroup 1 Group A			

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

2/ The LTPD for solderability tests applies to the number of leads inspected except in no case shall less than three devices be used to provide the number of leads required.

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TABLE VI. Group C inspection.

Inspection	Method	MIL-STD-883 details	LTPD
Subgroup 1			
Thermal shock	1010	Test condition A_1, except the minimum time for steps 1 and 3 shall be 2 hours.	10
Moisture resistance	1004		10
Hermetic seal	1014	Test condition A_1 for fine leaks (leak testing 2 hours after pressurization is acceptable); test condition C for gross leaks, except that leak indicator fluid shall be maintained at +100°C ±5°C.	10
External visual	2009		10
End points: Subgroup 1 Group A			
Subgroup 2			
Shock	2002	Non-operating; 500G; t = 1 ms; 5 blows in each orientation	10
Constant acceleration	2001	Non-operating; 5000 G; in each orientation	10
End points: Subgroup 1 Group A			
Subgroup 3			
Salt atmosphere	1009		15
Subgroup 4 1/			
Steady-state operation life	1005	T = +85°C Test condition B 1000 hours	λ=7
End points: Subgroup 1 Group A			

See footnote at end of table.

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TABLE VI. Group C inspection - Continued.

Inspection	Method	MIL-STD-883 details	LTPD
Subgroup 5			
Nuclear survivability	1019 1017		15
End points:			
Subgroup 1 Group A			

1/ Failed devices from the same inspection lot may be used for all subgroups when electrical or optical end-point measurements are not required.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - AV
Navy - AS, EC
Air Force - 17, 19, 70, 71, 80, 84, 90, 99
DLA - ES
NASA - NA

User activities:
Air Force - 13, 14

Preparing activity:
Army - CR

Agent:
DLA-ES

(Project 6030-0029)

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